

SQL

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ALTER TABLE student_details
ADD COLUMN Department VARCHAR(100);

UPDATE student_details
SET Department = "CSE"
WHERE Sno =1

UPDATE student_details
SET Department = "IT"
WHERE Sno =2

UPDATE student_details
SET Department = "ECE"
WHERE Sno =3

UPDATE student_details
SET Department = "AIDS"
WHERE Sno =4

-- dql -> data query language

SELECT * FROM student_details;

SELECT Name , Phone AS Mobile_Number FROM student_details;

SELECT * FROM student_details
WHERE Name = "Karthika" LIMIT 1;

SELECT * FROM student_details
WHERE Department IN ("CSE" , "IT", "ECE");

SELECT DISTINCT Name FROM student_details;

CREATE TABLE product(
    id INT ,
    product_name VARCHAR(200),
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    price FLOAT,
    stock INT
);

INSERT INTO product VALUES(1, "Coffee Mug", 40.50, 100),
(2, "Laptop Bag", 55, 25), (3, "mouse", 45, 400), (4, "Tshirt", 80, 20),
(5, "Remote", 90, 40);

INSERT INTO product VALUES(6, "Coffee", 50.50, 100);

SELECT * FROM product

SELECT * FROM product
WHERE price BETWEEN 40 AND 60;

SELECT * FROM product
WHERE product_name BETWEEN 'C' AND 'M';

-- like
SELECT * FROM product
WHERE product_name LIKE "c%"

SELECT * FROM product
WHERE product_name LIKE "%e"

SELECT * FROM product
WHERE product_name LIKE "%Bag%"

SELECT * FROM product
WHERE product_name LIKE "____e%"

-- dcl -> data control language
-- used to control user access and permissions in the database

CREATE USER 'karthika_fiit'@'localhost' IDENTIFIED BY '123';

GRANT SELECT , INSERT, UPDATE ON karthika.product TO
'karthika_fiit'@'localhost';

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REVOKE SELECT , INSERT, UPDATE ON karthika.product FROM
'karthika_fiit'@'localhost';
GRANT ALL PRIVILEGES ON karthika.* TO 'karthika_fiit'@'localhost';

FLUSH PRIVILEGES;

SHOW GRANTS FOR 'karthika_fiit'@'localhost';

REVOKE SELECT ON karthika.* FROM 'karthika_fiit'@'localhost';

REVOKE ALL PRIVILEGES ON karthika.* FROM 'karthika_fiit'@'localhost';

DROP USER 'karthika_fiit'@'localhost';

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-- TCL -> transaction control language
--used to manage transactions in a database to maintain data consistency and
data integrity

START TRANSACTION;

UPDATE product
SET price = price*100
WHERE id=2;

ROLLBACK; -- restores the database to the last committed state

COMMIT;-- permanently stores data

-- savepoint

START TRANSACTION

UPDATE product
SET price = price/2
WHERE id=4;

SAVEPOINT sp1;

UPDATE product
SET price = price *10
WHERE id=2;

ROLLBACK TO sp1;

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COMMIT;  
  
SELECT @@autocommit;  
  
SET autocommit = 0;  
  
SET autocommit = 1;
```

Theory Notes

1. SQL Operators

- **Arithmetic** → +, -, *, /, %.
- **Comparison** → =, != or <>, >, <, >=, <=.
- **Logical** → AND, OR, NOT.
- **Special** → BETWEEN, IN, LIKE, IS NULL.

2. SQL Commands

- **DDL (Data Definition Language)** → CREATE, ALTER, DROP, TRUNCATE.
- **DML (Data Manipulation Language)** → INSERT, UPDATE, DELETE.
- **DQL (Data Query Language)** → SELECT.
- **DCL (Data Control Language)** → GRANT, REVOKE.
- **TCL (Transaction Control Language)** → COMMIT, ROLLBACK, SAVEPOINT.

3. Key Concepts

- **ALTER TABLE** → add, modify, rename, or drop columns.
- **UPDATE** → modify existing records.
- **SELECT** → query records with filters (WHERE, IN, BETWEEN, LIKE).
- **DISTINCT** → remove duplicates.
- **Privileges** → GRANT and REVOKE control user access.
- **Transactions** → ensure data consistency with COMMIT, ROLLBACK, SAVEPOINT.

Coding Round Questions

- Add column → Write a query to add a column email to student_details.
- Update record → Update department of student with Sno=2 to EEE.
- Select distinct → Select distinct names from student_details.
- Between operator → Find products priced between 100 and 500.
- Like operator → Find products whose names start with Lap.
- Grant privileges → Grant SELECT and INSERT privileges on product table to a user.
- Rollback transaction → Demonstrate rollback after updating a product price.